

# Computer Vision Lab Exam — MCQ Cheat Sheet

150 exam-style MCQs based on the hard-question/code cheat sheet prepared in this conversation. These are practice predictions, not guaranteed exam questions.

**Must-remember:** 3 points → affine | 4 points → homography | adaptive threshold → uneven lighting | Otsu → automatic threshold | opening → remove small noise | closing → fill small gaps | watershed → touching objects | Canny → edges | HoughLinesP → lines | HoughCircles → circles | SIFT + ratio test → matching | RANSAC → outliers | warpPerspective → homography | bitwise\_and + mask → ROI.

## OpenCV Basics

### 1. Which function reads an image from disk?

- A. cv2.imread
- B. cv2.readImage
- C. cv2.load
- D. cv2.open

### 2. cv2.imread normally returns color channels in which order?

- A. RGB
- B. BGR
- C. HSV
- D. GRAY

### 3. Which conversion turns BGR into grayscale?

- A. cv2.COLOR\_BGR2GRAY
- B. cv2.COLOR\_RGB2GRAY
- C. cv2.COLOR\_BGR2HSV
- D. cv2.GRAY2BGR

### 4. What is the usual 8-bit grayscale range?

- A. 0-1
- B. 0-100
- C. 0-255
- D. 1-256

### 5. Which function changes an image size?

- A. cv2.resize
- B. cv2.scale
- C. cv2.changeSize
- D. cv2.reshapeImage

### 6. Which function creates a rotation matrix around a point?

- A. cv2.getRotationMatrix2D
- B. cv2.rotateMatrix
- C. cv2.rotationMatrix
- D. cv2.makeRotation

### 7. Which function applies a 2D affine matrix to an image?

- A. cv2.warpAffine
- B. cv2.warpPerspective
- C. cv2.applyAffineOnly
- D. cv2.affineWarp

### 8. Why convert BGR to RGB before normal color display with Matplotlib?

- A. Matplotlib normally expects RGB order
- B. It increases resolution
- C. It removes noise
- D. It makes the image grayscale

### 9. Which statement about NumPy image slicing is correct?

- A. It can crop/select rows and columns
- B. It detects circles
- C. It computes homography
- D. It creates SIFT descriptors

### 10. For img[100:400,100:500], the first range refers to:

- A. Columns
- B. Rows
- C. Channels
- D. Color values

## Histogram and Enhancement

**11. Which function computes a grayscale histogram in OpenCV?**

- A. cv2.calcHist
- B. cv2.histogram
- C. cv2.getHistogram
- D. cv2.makeHist

**12. Histogram equalization mainly improves:**

- A. Contrast
- B. Image dimensions
- C. FPS
- D. Number of channels

**13. Which function performs grayscale histogram equalization?**

- A. cv2.equalizeHist
- B. cv2.histEqual
- C. cv2.equalize
- D. cv2.histNormalize

**14. A histogram concentrated on the left usually indicates an image that is:**

- A. Bright
- B. Dark
- C. Highly saturated
- D. Rotated

**15. A histogram concentrated on the right usually indicates an image that is:**

- A. Dark
- B. Bright
- C. Blurred
- D. Binary only

**16. Which transform is useful for enhancing dark details?**

- A. Log transform
- B. Translation
- C. Rotation
- D. Cropping

**17. For  $\gamma < 1$ , the image is generally:**

- A. Brightened
- B. Darkened
- C. Rotated
- D. Binarized

**18. For  $\gamma > 1$ , the image is generally:**

- A. Brightened
- B. Darkened
- C. Segmented by color
- D. Warped

**19. Why use float values for a log transform?**

- A. To safely perform floating-point arithmetic
- B. To change BGR to RGB
- C. To detect contours
- D. To increase FPS

**20. Which enhancement changes intensity mapping without changing image geometry?**

- A. Gamma correction
- B. Rotation
- C. Translation
- D. Perspective warp

## Thresholding and ROI

**21. Which function performs binary thresholding?**

- A. cv2.threshold
- B. cv2.Canny
- C. cv2.findContours
- D. cv2.HoughLinesP

**22. Which threshold method chooses a threshold automatically from the histogram?**

- A. Otsu
- B. SIFT
- C. Hough
- D. Watershed

**23. When using Otsu, the threshold argument is commonly set to:**

- A. 0
- B. 127
- C. 255
- D. -1

**24. Which method handles spatially varying illumination better?**

- A. Adaptive thresholding
- B. Fixed global threshold
- C. Affine transform
- D. SIFT

**25. adaptiveThreshold blockSize should normally be:**

- A. Even
- B. Odd
- C. Negative
- D. A floating point number

**26. In THRESH\_BINARY, a pixel above the threshold is commonly assigned:**

- A. 0
- B. 1
- C. 127
- D. 255

**27. Which function creates a polygon mask?**

- A. cv2.fillPoly
- B. cv2.makeMask
- C. cv2.roiPolygon
- D. cv2.drawROI

**28. Which value commonly represents the selected region in an 8-bit mask?**

- A. 0
- B. 1
- C. 128
- D. 255

**29. How do you apply a grayscale ROI mask in the cheat-sheet code?**

- A. cv2.bitwise\_and(gray,gray,mask=mask)
- B. cv2.equalizeHist(gray,mask)
- C. cv2.Canny(gray,mask)
- D. cv2.resize(gray,mask)

**30. Creating a mask without applying it means you have:**

- A. Only defined the ROI mask
- B. Already extracted the ROI
- C. Detected edges
- D. Found contours

## Filtering, Edges and Contours

**31. Which function detects edges with the Canny algorithm?**

- A. cv2.Canny
- B. cv2.EdgeDetect
- C. cv2.findEdges
- D. cv2.HoughEdges

**32. Why blur before Canny?**

- A. To reduce noise
- B. To add color
- C. To create keypoints
- D. To resize the image

**33. Which filter is especially useful for salt-and-pepper noise?**

- A. Median blur
- B. Hough blur
- C. SIFT blur
- D. Otsu blur

**34. Which function finds object boundaries after segmentation?**

- A. cv2.findContours
- B. cv2.findEdgesOnly
- C. cv2.detectBoundary
- D. cv2.boundaries

**35. What does cv2.contourArea return?**

- A. Area of a contour
- B. Perimeter only
- C. Circle radius
- D. Line angle

**36. What does cv2.boundingRect return?**

- A. x,y,width,height
- B. x1,y1,x2,y2 only
- C. center,radius
- D. angle,scale

**37. Which contour retrieval mode keeps only external contours?**

- A. RETR\_EXTERNAL
- B. RETR\_INTERNAL
- C. RETR\_OUTER\_ONLY
- D. RETR\_SIMPLE

**38. A simple way to count segmented objects is:**

- A. Count contours after filtering by area
- B. Count histogram bins
- C. Count Canny thresholds
- D. Count image channels

**39. Why filter contours by area?**

- A. To reject tiny unwanted blobs
- B. To rotate objects
- C. To calculate RGB values
- D. To find homography

**40. Which operation is mainly intensity/edge based rather than geometric?**

- A. Canny
- B. Rotation
- C. Translation
- D. Cropping

## Segmentation, Morphology and Watershed

**41. In K-means image segmentation, K represents:**

- A. Number of clusters
- B. Kernel size
- C. Number of channels
- D. Number of images

**42. Why reshape RGB pixels to (-1,3) for K-means?**

- A. Each pixel becomes one 3-value sample
- B. It rotates the image
- C. It creates contours
- D. It changes RGB to HSV

**43. Why use float32 pixels for cv2.kmeans?**

- A. It expects numerical floating-point samples
- B. It detects edges better
- C. It creates a mask
- D. It saves memory only

**44. Opening consists of:**

- A. Erosion then dilation
- B. Dilation then erosion
- C. Blur then threshold
- D. Threshold then Canny

**45. Closing consists of:**

- A. Dilation then erosion
- B. Erosion then dilation
- C. Canny then Hough
- D. SIFT then RANSAC

**46. Opening is commonly used to:**

- A. Remove small noise
- B. Fill small holes
- C. Detect lines
- D. Estimate homography

**47. Closing is commonly used to:**

- A. Fill small gaps/holes
- B. Remove every object
- C. Detect SIFT
- D. Calculate a histogram

**48. Watershed is particularly useful for:**

- A. Separating touching objects
- B. Changing brightness
- C. Finding circles only
- D. Rotating images

**49. In cv2.watershed, pixels labeled -1 represent:**

- A. Region boundaries
- B. The first object
- C. Background only
- D. Invalid input

**50. Which operation commonly provides foreground markers for watershed?**

- A. Distance transform plus thresholding
- B. Histogram equalization only
- C. Hough circles
- D. Gamma correction

## Hough Transform and Lane Detection

**51. Which method detects straight lines?**

- A. Hough line transform
- B. Hough circle transform
- C. SIFT
- D. Watershed

**52. Which function detects line segments with the probabilistic Hough transform?**

- A. cv2.HoughLinesP
- B. cv2.HoughCircles
- C. cv2.findLines
- D. cv2.LineTransform

**53. In HoughLinesP, minLineLength controls:**

- A. Minimum accepted line length
- B. Minimum image width
- C. Minimum brightness
- D. Minimum circle radius

**54. In HoughLinesP, maxLineGap controls:**

- A. Maximum gap for joining line segments
- B. Maximum image height
- C. Maximum edge threshold
- D. Maximum FPS

**55. np.pi/180 in a Hough call represents:**

- A. One degree in radians
- B. One pixel
- C. One channel
- D. One FPS

**56. Which method detects circles?**

- A. cv2.HoughCircles
- B. cv2.HoughLinesP
- C. cv2.findContours only
- D. cv2.SIFT\_create

**57. In HoughCircles, param2 is mainly related to:**

- A. Accumulator threshold for accepting circles
- B. Image width
- C. Number of channels
- D. Gamma

**58. A typical lane pipeline is:**

- A. Grayscale → blur → Canny → ROI → HoughLinesP
- B. SIFT → gamma → circles
- C. Otsu → MRI fusion → SIFT
- D. Wavelet → histogram → rotation

**59. Why use an ROI in lane detection?**

- A. To focus on the road/lane region
- B. To increase color depth
- C. To calculate histogram
- D. To detect SIFT

**60. If HoughLinesP detects too many short lines, a useful adjustment is:**

- A. Increase minLineLength
- B. Set gamma to zero
- C. Use addWeighted
- D. Remove grayscale conversion

## Affine and Homography

**61. How many point pairs are minimally needed for a 2D affine transform?**

- A. 2
- B. 3
- C. 4
- D. 8

**62. How many point pairs are minimally needed for a homography?**

- A. 2
- B. 3
- C. 4
- D. 8

**63. Which function calculates an affine matrix from three point pairs?**

- A. cv2.getAffineTransform
- B. cv2.findHomography
- C. cv2.warpAffine
- D. cv2.getPerspective

**64. Which function calculates a homography?**

- A. cv2.findHomography
- B. cv2.getAffineTransform
- C. cv2.warpAffine
- D. cv2.HoughLinesP

**65. Which function applies an affine transform to an image?**

- A. cv2.warpAffine
- B. cv2.warpPerspective
- C. cv2.applyAffine
- D. cv2.transformAffineImage

**66. Which function applies a perspective transform using H?**

- A. cv2.warpPerspective
- B. cv2.warpAffine
- C. cv2.applyHomographyImage
- D. cv2.perspectiveWarpOnly

**67. Which transform is needed to model perspective distortion between planar views?**

- A. Homography
- B. Median filtering
- C. Histogram equalization
- D. Opening

**68. Which statement is the best exam shortcut?**

- A. 3 points → affine; 4 points → homography
- B. 3 points → homography; 4 → affine
- C. Both need 8 points
- D. Neither needs point correspondences

**69. RANSAC is commonly used with homography to:**

- A. Reject outlier correspondences
- B. Increase brightness
- C. Detect circles
- D. Resize images

**70. A document photographed at an angle is a classic use case for:**

- A. Homography and perspective warp
- B. Opening only
- C. Canny only
- D. K-means only

## **SIFT, Matching and Panorama**

**71. What does SIFT provide for matching?**

- A. Keypoints and descriptors
- B. Only a histogram
- C. Only contours
- D. Only line segments

**72. Which call creates a SIFT detector?**

- A. cv2.SIFT\_create()
- B. cv2.createSIFT()
- C. cv2.SIFT()
- D. cv2.makeSIFT()

**73. Why are descriptors needed?**

- A. To compare local feature appearance
- B. To resize images
- C. To create binary masks
- D. To calculate FPS

**74. Which matcher call supports the two-nearest-neighbor ratio test?**

- A. BFMatcher().knnMatch(...,k=2)
- B. HoughLinesP(...,k=2)
- C. findContours(...,k=2)
- D. equalizeHist(...,k=2)

**75. Why compare the best and second-best SIFT matches?**

- A. To reject ambiguous matches
- B. To find two circles
- C. To calculate two histograms
- D. To double image size

**76. A common ratio threshold in the cheat-sheet code is:**

- A. 0.25
- B. 0.50
- C. 0.75
- D. 1.50

**77. A smaller descriptor distance generally means:**

- A. A more similar descriptor match
- B. A larger object
- C. A brighter pixel
- D. A longer line

**78. After good SIFT matches, which model maps one planar view to another?**

- A. Homography
- B. Gamma
- C. Histogram
- D. Watershed

**79. A basic panorama pipeline is:**

- A. SIFT → matching → ratio test → homography → warp
- B. Otsu → circles → gamma → K-means
- C. Opening → closing → video
- D. Histogram → FPS → threshold

**80. Why does a panorama need overlap between images?**

- A. Overlap supplies common features for matching
- B. Overlap increases grayscale range
- C. Overlap is required for gamma
- D. Overlap creates circles

## Medical Imaging, Fusion and Wavelets

**81. Which function performs weighted CT/MRI fusion in the cheat sheet?**

- A. cv2.addWeighted
- B. cv2.addFusion
- C. cv2.fuseImages
- D. cv2.weightedBlend

**82. Which call gives 70% CT and 30% MRI?**

- A. cv2.addWeighted(ct,0.7,mri,0.3,0)
- B. cv2.addWeighted(ct,0.3,mri,0.7,0)
- C. cv2.threshold(ct,mri,0.7)
- D. np.maximum(ct,mri,0.7)

**83. Before weighted fusion, CT and MRI should have:**

- A. Compatible dimensions and types
- B. Different dimensions
- C. Different FPS
- D. Only RGB channels

**84. What does np.maximum(a,b) do pixelwise for same-sized grayscale images?**

- A. Selects the larger intensity
- B. Averages intensities
- C. Subtracts images
- D. Rotates images

**85. Which library is used for the wavelet example?**

- A. PyWavelets (pywt)
- B. Pandas
- C. Flask
- D. TensorFlow only

**86. In a 2D Haar decomposition, LL is:**

- A. Low-frequency approximation
- B. Horizontal detail only
- C. Vertical detail only
- D. Diagonal detail only

**87. Which components contain high-frequency detail?**

- A. LH, HL, HH
- B. LL only
- C. RGB only
- D. XYZ only

**88. A simple wavelet anomaly detector can flag:**

- A. Large detail coefficients
- B. Only bright pixels
- C. Only image borders
- D. FPS changes

**89. Which task is especially suited to X-ray contrast enhancement?**

- A. Histogram equalization or log/gamma enhancement
- B. Hough circles only
- C. VideoWriter
- D. pointPolygonTest

**90. Which thresholding approach can be useful on an enhanced X-ray with uneven illumination?**

- A. Adaptive thresholding
- B. Only fixed 127 threshold
- C. HoughLinesP
- D. SIFT matching

## Video, Statistics and Exam Traps

**91. Which class reads video frames?**

- A. cv2.VideoCapture
- B. cv2.VideoReader
- C. cv2.FrameReader
- D. cv2.readVideo

**92. What does cap.read() return?**

- A. ret and frame
- B. frame and FPS
- C. image and mask
- D. x and y

**93. When ret is False, the normal response is to:**

- A. Stop because no frame was read
- B. Run SIFT
- C. Increase gamma
- D. Run HoughCircles

**94. Which call releases a video capture?**

- A. cap.release()
- B. cap.closeVideo()
- C. cv2.releaseVideo()
- D. cap.stop()

**95. Which call closes OpenCV windows?**

- A. cv2.destroyAllWindows()
- B. cv2.closeWindows()
- C. cv2.releaseWindows()
- D. plt.destroyVideo()

**96. Which key is used to quit in the cheat-sheet video loop?**

- A. q
- B. x
- C. e
- D. z

**97. Simple frame-difference motion detection uses:**

- A. cv2.absdiff between consecutive frames
- B. Histogram equalization only
- C. Hough circles
- D. SIFT only

**98. Why threshold the frame difference?**

- A. To create a binary motion mask
- B. To estimate homography
- C. To create descriptors
- D. To increase FPS

**99. Which NumPy function gives the mean intensity?**

- A. np.mean
- B. np.avgImage
- C. np.meanValue
- D. np.averageOnly

**100. Which NumPy function gives standard deviation?**

- A. np.std
- B. np.standard
- C. np.dev
- D. np.stdevImage

## Twisted Exam-Style Questions

- 101. You made a polygon mask but the outside is not black. What is most likely missing?**
- A. Applying the mask with bitwise\_and
  - B. Calling equalizeHist twice
  - C. Using SIFT
  - D. Calling VideoWriter
- 102. Lighting changes from one side of an image to another. Which segmentation choice is safer?**
- A. Adaptive threshold
  - B. Fixed global threshold only
  - C. HoughCircles
  - D. Homography
- 103. Two coins touch after thresholding. Which method is designed to separate them?**
- A. Watershed
  - B. Gamma correction
  - C. Histogram equalization
  - D. HSV threshold
- 104. Many SIFT matches are wrong. Which combination is strongest?**
- A. Ratio test + RANSAC homography
  - B. Global threshold + closing
  - C. Gamma + histogram
  - D. Hough circles + opening
- 105. A road image contains buildings and wires. Why restrict Hough detection to a lane ROI?**
- A. To reduce irrelevant line detections
  - B. To increase bit depth
  - C. To change BGR to RGB
  - D. To create descriptors
- 106. CT and MRI images have different dimensions. What should happen before direct addWeighted fusion?**
- A. Resize/alignment to compatible dimensions
  - B. Run HoughCircles
  - C. Convert to video
  - D. Use pointPolygonTest
- 107. A perspective warp leaves black areas around the result. Why?**
- A. The output canvas contains locations with no source pixels
  - B. SIFT always creates black pixels
  - C. Canny forces black borders
  - D. RANSAC deletes pixels
- 108. Four corner correspondences are given to rectify a tilted image. Which pair is central?**
- A. findHomography + warpPerspective
  - B. findContours + equalizeHist
  - C. Canny + HoughCircles
  - D. K-means + opening
- 109. A binary image has isolated tiny foreground dots. Which is usually better?**
- A. Opening
  - B. Closing
  - C. Homography
  - D. SIFT
- 110. A binary object has tiny breaks that should be connected. Which is usually better?**
- A. Closing
  - B. Opening
  - C. Canny
  - D. Hough

## What Should I Use?

- 111. Question says: “enhance contrast.” Best first choice?**
- A. Histogram equalization
  - B. HoughCircles
  - C. Watershed
  - D. VideoCapture
- 112. Question says: “image is very dark.” Best candidates?**
- A. Log/gamma/equalization
  - B. HoughLinesP only
  - C. pointPolygonTest
  - D. SIFT only

**113. Question says: “uneven illumination.”**

- A. Adaptive threshold
- B. Fixed threshold only
- C. Affine transform
- D. Hough circles

**114. Question says: “automatic threshold.”**

- A. Otsu
- B. SIFT
- C. Hough
- D. Gamma

**115. Question says: “detect edges.”**

- A. Canny
- B. K-means
- C. Homography
- D. addWeighted

**116. Question says: “detect straight lane markings.”**

- A. HoughLinesP
- B. HoughCircles
- C. SIFT
- D. Watershed

**117. Question says: “detect circular objects.”**

- A. HoughCircles
- B. HoughLinesP
- C. Canny only
- D. equalizeHist

**118. Question says: “divide pixels into K groups.”**

- A. K-means
- B. RANSAC
- C. SIFT
- D. Watershed only

**119. Question says: “recognize the same planar object in another image.”**

- A. SIFT + matching + homography
- B. Otsu only
- C. Opening only
- D. Gamma only

**120. Question says: “stitch overlapping images.”**

- A. Feature matching + homography + warping
- B. Thresholding only
- C. Closing only
- D. Mean intensity only

## Parameter and Code Traps

**121. In cv2.Canny(gray,50,150), the two numbers are mainly:**

- A. Lower and upper edge thresholds
- B. Width and height
- C. Min/max radius
- D. HSV lower/upper

**122. In cv2.HoughLinesP(edges,1,np.pi/180,50,...), the 1 is the:**

- A. Distance resolution in pixels
- B. Number of channels
- C. Image width
- D. FPS

**123. In cv2.HoughLinesP, np.pi/180 is the:**

- A. Angle resolution in radians
- B. Line width
- C. Circle radius
- D. Gamma

**124. In cv2.HoughCircles, minDist controls:**

- A. Minimum distance between circle centers
- B. Minimum line length
- C. Minimum image width
- D. Minimum threshold

**125. In cv2.HoughCircles, minRadius controls:**

- A. Smallest allowed circle radius
- B. Smallest line gap
- C. Smallest contour area
- D. Smallest image dimension

**126. In `cv2.findHomography(src,dst,cv2.RANSAC,5.0)`, 5.0 is commonly:**

- A. RANSAC reprojection threshold
- B. Image width
- C. Number of matches
- D. Gamma

**127. In `cv2.addWeighted(a,alpha,b,beta,gamma)`, the last gamma is:**

- A. Scalar added to the weighted sum
- B. Gamma correction exponent
- C. HSV hue
- D. Homography parameter

**128. In `cv2.resize(img,(500,400))`, the tuple is:**

- A. New width, new height
- B. New height, new width
- C. Rows, columns
- D. x,y coordinates

**129. In `np.zeros(gray.shape,dtype=np.uint8)`, the dtype is used because a mask is:**

- A. An 8-bit image
- B. A float descriptor
- C. A video
- D. A homography

**130. In `cv2.fillPoly(mask,[points],255)`, 255 means:**

- A. Fill the polygon with white/selected pixels
- B. Use a threshold of 255
- C. Rotate by 255 degrees
- D. Use 255 points

## Hard Combined Questions

**131. Best pipeline for lane detection under noise?**

- A. Gray → Gaussian blur → Canny → ROI → HoughLinesP
- B. SIFT → Otsu → fusion
- C. K-means → circles → gamma
- D. Watershed → video writer

**132. Best pipeline for recognizing a template under perspective change?**

- A. SIFT → ratio test → RANSAC homography → perspective transform
- B. Otsu → opening → closing
- C. Canny → HoughCircles only
- D. Histogram → gamma only

**133. Best pipeline for separating touching bright objects in a dark image?**

- A. Threshold → morphology → distance transform → watershed
- B. SIFT → homography
- C. HSV → HoughLinesP
- D. Gamma → VideoCapture

**134. Best simple approach for a red object on a varied background?**

- A. HSV threshold + mask
- B. Grayscale only
- C. Homography only
- D. Wavelet only

**135. If an image has salt-and-pepper noise before edge detection, use:**

- A. Median blur then Canny
- B. HoughCircles then SIFT
- C. Gamma then VideoWriter
- D. Homography then Otsu

**136. If contours detect hundreds of tiny noise blobs, first simple fix?**

- A. Filter by contour area and/or use opening
- B. Use more SIFT features
- C. Increase panorama width
- D. Use addWeighted

**137. If SIFT has fewer than four good matches, homography estimation is:**

- A. Not reliably possible with the basic minimum requirement
- B. Guaranteed accurate
- C. Automatically replaced by affine
- D. Unnecessary

**138. If the exam asks for 3 matching points and “shear/rotation/translation,” choose:**

- A. Affine transformation
- B. Homography
- C. Watershed
- D. Hough

**139. If the exam asks for 4 matching corners and “perspective,” choose:**

- A. Homography
- B. Median filter
- C. Opening
- D. Canny

**140. If the exam asks to save processed frames to an MP4, use:**

- A. cv2.VideoWriter
- B. cv2.VideoCapture only
- C. cv2.calcHist
- D. cv2.findHomography

## **Rapid-Fire Viva-to-MCQ Theory**

**141. Which transformation can include translation, rotation, scaling and shear?**

- A. Affine
- B. Histogram equalization
- C. Watershed
- D. HSV threshold

**142. Which transformation can represent perspective effects?**

- A. Projective/homography
- B. Median blur
- C. Opening
- D. Canny

**143. Which method is robust to some incorrect correspondences?**

- A. RANSAC
- B. Otsu
- C. K-means
- D. Gamma

**144. Which method is invariant-ish to scale and rotation and useful for local feature matching?**

- A. SIFT
- B. Otsu
- C. HoughCircles
- D. Opening

**145. Which method is unsupervised clustering?**

- A. K-means
- B. Canny
- C. RANSAC
- D. SIFT

**146. Which operation is erosion followed by dilation?**

- A. Opening
- B. Closing
- C. Watershed
- D. Homography

**147. Which operation is dilation followed by erosion?**

- A. Closing
- B. Opening
- C. Canny
- D. SIFT

**148. Which segmentation method uses local neighborhoods for threshold selection?**

- A. Adaptive thresholding
- B. Fixed threshold
- C. Homography
- D. SIFT

**149. Which transform converts a point from one planar view using H?**

- A. cv2.perspectiveTransform
- B. cv2.pointPolygonTest
- C. cv2.equalizeHist
- D. cv2.calcHist

**150. Which function checks whether a point is inside a polygon?**

- A. cv2.pointPolygonTest
- B. cv2.findContours
- C. cv2.fillPoly
- D. cv2.inRange

## Answer Key

1. A 2. B 3. A 4. C 5. A 6. A 7. A 8. A 9. A 10. B 11. A 12. A 13. A 14. B 15. B  
16. A 17. A 18. B 19. A 20. A 21. A 22. A 23. A 24. A 25. B 26. D 27. A 28. D 29. A 30. A  
31. A 32. A 33. A 34. A 35. A 36. A 37. A 38. A 39. A 40. A 41. A 42. A 43. A 44. A 45. A  
46. A 47. A 48. A 49. A 50. A 51. A 52. A 53. A 54. A 55. A 56. A 57. A 58. A 59. A 60. A  
61. B 62. C 63. A 64. A 65. A 66. A 67. A 68. A 69. A 70. A 71. A 72. A 73. A 74. A 75. A  
76. C 77. A 78. A 79. A 80. A 81. A 82. A 83. A 84. A 85. A 86. A 87. A 88. A 89. A 90. A  
91. A 92. A 93. A 94. A 95. A 96. A 97. A 98. A 99. A 100. A 101. A 102. A 103. A 104. A 105. A  
106. A 107. A 108. A 109. A 110. A 111. A 112. A 113. A 114. A 115. A 116. A 117. A 118. A 119. A 120. A  
121. A 122. A 123. A 124. A 125. A 126. A 127. A 128. A 129. A 130. A 131. A 132. A 133. A 134. A 135. A  
136. A 137. A 138. A 139. A 140. A 141. A 142. A 143. A 144. A 145. A 146. A 147. A 148. A 149. A 150. A

**Exam warning:** Your teacher may twist the wording or combine multiple techniques. If a question describes a goal, identify the goal first, then choose the matching pipeline. These MCQs are a study aid, not a prediction of exact questions.